PROJECT SYNOPSIS

Campus Event Manager – A Flutter-based College Event Management Application

(Event Listings | Student Registrations | Push Notifications)

Course: Flutter Hybrid Mobile Application Development

1. Introduction

College campuses host a large number of events throughout the academic year — technical fests, workshops, seminars, cultural programs, and club activities. However, students often miss out on these events due to scattered information across notice boards, WhatsApp groups, and social media, while event organizers struggle to manage registrations and communicate reminders efficiently. Campus Event Manager is a Flutter-based mobile application designed to centralize event discovery, registration, and notification for students and organizers on a single platform.

2. Problem Statement

- Event information is scattered across multiple informal channels, causing students to miss events.
- There is no structured system for students to register for events digitally.
- Organizers have no easy way to track who has registered for an event.
- Students do not receive timely reminders/notifications about events they are interested in.

3. Proposed System

Campus Event Manager allows organizers/admins to create and publish events with details such as date, time, venue, and description. Students can browse events, view venue locations on Google Maps, register through a validated form, and receive push notifications as reminders. The app uses Firebase for authentication and cloud data, a REST API backend for event data, local storage for offline access to registered events, and Firebase Cloud Messaging for notifications.

4. Objectives

1. To design and develop a fully functional, responsive Flutter mobile application for event management.
2. To implement secure user authentication using Firebase.
3. To integrate REST APIs for dynamic event data handling.
4. To implement local storage for offline access to a student's registered events.
5. To integrate Google Maps for displaying event venue locations.
6. To implement push notifications for event reminders using Firebase Cloud Messaging.
7. To deliver a professional, well-documented, and demo-ready mobile application.

5. Target Users

- College/University students (primary users, browse and register for events).
- Event organizers / club coordinators / admin (create and manage events).
- College administration (monitor overall event participation).

6. Scope of the Project

The application covers event listing and discovery, event creation/management, student registration, and notification reminders, built on top of authentication and profile management. The scope is limited to a functional MVP (Minimum Viable Product); features such as online payment for paid events or QR-code-based attendance are out of scope.

7. System Modules / Screens

Screen No.	Screen Name	Purpose / Key Feature
1	Splash / Login Screen	Firebase Authentication (email/password login & signup) with input validation and loading indicator
2	Home / Event List Screen	Displays list of upcoming college events fetched via REST API/Firestore; search & category filter; loading & error states
3	Event Detail Screen	Full event details (date, time, venue, description); Google Map showing venue location; Register button
4	Add/Edit Event Screen (Admin/Organizer)	Form with validation to create or update an event (CRUD - Create/Update), including image/banner upload
5	Registration Form Screen	Form with validation for a student to register for an event; saves registration via REST API and local storage
6	My Registrations Screen	Shows events the student has registered for, stored locally (SQLite/Hive) with option to cancel (CRUD - Delete)
7	Notifications Screen	List of push notifications/reminders about registered or upcoming events (Firebase Cloud Messaging)
8	Profile & Settings Screen	User profile info, edit details, notification preferences, logout

8. Core Functional Requirements

- Minimum 6 screens with a responsive, professional UI/UX
- Navigation between screens (bottom navigation + stack navigation)
- Forms with input validation (event creation, registration, login/signup)
- CRUD operations (Create, Read, Update, Delete on events and registrations)
- Local storage using SQLite/Hive/Shared Preferences for registered events
- REST API integration with loading indicators and error handling
- Firebase Authentication, Cloud Firestore, Google Maps, Image Upload, and Push Notifications as bonus features

9. Technologies Used

Category	Technology / Tool
Frontend Framework	Flutter (Dart)
State Management	Provider / Riverpod
Local Storage	SQLite (sqflite) / Hive / Shared Preferences
Backend / REST API	Node.js + Express (or public REST API) with JSON responses for event data
Cloud Services	Firebase Authentication, Cloud Firestore, Firebase Cloud Messaging (push notifications)
Maps & Location	Google Maps Flutter Plugin, Geolocator (event venue display)
Version Control	Git & GitHub
Design Tool	Figma (for wireframes/UI mockups)

10. Hardware & Software Requirements

Requirement Type	Specification
Development OS	Windows / macOS / Linux
IDE	Android Studio / Visual Studio Code
SDK	Flutter SDK, Dart SDK
Minimum RAM	8 GB (recommended for smooth emulator performance)
Testing Device	Android/iOS smartphone or emulator (Android 8.0+ / iOS 12+)
Internet Connection	Required for REST API and Firebase services

11. Expected Output

A fully functional cross-platform (Android/iOS) mobile application where:

- Students can sign up/log in securely and browse a live list of campus events.
- Students can view event details with venue location on a map and register with a single validated form.
- Registered events are visible offline in the 'My Registrations' section and can be cancelled.
- Students receive push notifications reminding them of upcoming registered events.
- Organizers/admins can create, edit, and delete events with image banners in real time.
- A working APK is generated that can be installed and demonstrated on a physical device or emulator.

12. Advantages of the Proposed System

- Centralized, always up-to-date source of campus event information.
- Reduces manual effort in tracking event registrations.
- Improves student engagement through timely notifications.
- Location-based venue guidance reduces confusion on event day.
- Offline access ensures students can view their registrations without an active internet connection.

13. Future Scope

- QR-code based attendance marking at event venues.
- Integration of online payment for paid events.
- Analytics dashboard for organizers to track registration trends.
- In-app chat/discussion forum for event participants.

14. Conclusion

Campus Event Manager combines practical, everyday utility for students and organizers with comprehensive coverage of Flutter's core and advanced capabilities — including REST API integration, Firebase services, local storage, and Google Maps — resulting in a cohesive, real-world mobile application.